

An autoregressive model with time-varying coefficients for wind fields

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SUMMARY

In this article, an original Markov-switching autoregressive model is proposed to describe the space–time evolution of wind fields. At first, a non-observable process is introduced in order to model the motion of the meteorological structures. Then, conditionally to this process, the evolution of the wind fields is described using autoregressive models with time-varying coefficients. The proposed model is calibrated and validated on data in the North Atlantic. Copyright © 2005 John Wiley & Sons, Ltd.

KEY WORDS: space–time model; wind fields; Markov-switching autoregressive model

1. INTRODUCTION

Wind conditions have a determining influence on many human activities and physical phenomena. For instance, wind time series permit the evaluation of the power values produced by wind turbines (Brown *et al.*, 1984; Castino *et al.*, 1998) to predict the propagation of an oil-spill, to assess the profitability of a maritime line (Ailliot *et al.*, 2003) or to study coastal erosion. In order to understand and predict the evolution of these phenomena, it is useful to develop stochastic models for wind time series. Such models can be used to make short-term forecasts, to perform Monte Carlo studies (Ailliot *et al.*, 2003) or to assimilate in-situ measurements in numerical forecasts (Malmberg *et al.*, 2005).

For some applications, it is sufficient to have a model which describes the evolution of the wind in a fixed location, and different models have been proposed (see Brown *et al.*, 1984; Toll, 1997; Ailliot *et al.*, 2003; Monbet *et al.*, 2005). For other applications, it is necessary to know the wind conditions in several points simultaneously, and a space–time model is useful. The few models which have been proposed in the literature are based on Lagrangian or Eulerian methods. In the first approach, one follows the space–time evolution of some entities, along the track of storms, for example. It has been used, for example, by Casson and Coles (1998) to predict extreme wind speeds in hurricanes or by

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Boukhanovsky *et al.* (2003b) for storms in the Barents sea. In the second approach, one models directly the evolution of the process at a fixed set of locations. For most of the applications, it is necessary to know the wind conditions on a fixed spatial grid, and the Eulerian approach is then more appropriate. We focus on this approach in this article.

Boukhanovsky *et al.* (2003a) and Malmberg *et al.* (2005) proposed linear autoregressive models to describe time series of wind fields. In these two papers, a principal component analysis is first used in order to reduce the dimension of the observations. However, this type of model cannot reproduce some features of the wind fields, and in particular the motions of the meteorological structures. In this article, we propose an original model, in which these motions are introduced as a hidden Markov chain. Conditionally to this hidden process, the evolution of the wind fields is modeled using autoregressive models with time-varying coefficients.

In the first part, we present the data and the model. Then, in the second part, we briefly discuss the problem of parameter estimation, and finally, the last part is devoted to model validation.

2. MODEL DESCRIPTION

2.1. The data

In this article, we have used hindcast data produced by ECMWF (European Center for Medium Range Weather Forecast). This dataset describes the wind conditions at 10 m above sea-level for the period 1992–2003, on a regular grid with a resolution of 1.125 degrees both in latitude and longitude and a time-step of 6 h. This dataset is available everywhere on earth, and we have extracted an area which is located between latitudes 45 N and 50 N and longitudes 23 W and 31 W (see Figure 1). Its dimension is about 600 km per 600 km, and it contains $N = 35$ points of the grid of the hindcast model. These points will be denoted by $R_0 = (r_1, \dots, r_N)$. They have been ordered in the following way: $r_i = (x_i, y_i) < r_j = (x_j, y_j)$ if $x_i < x_j$ or $x_i = x_j$ and $y_i < y_j$. With this convention, the point r_1 is located at the south-west of the area R_0 , r_5 at the north-west, r_{31} at the south-east and r_{35} at the north-east.

We will denote $Z_t(R_0) = (u_t(r_1), u_t(r_2), \dots, u_t(r_N), v_t(r_1), v_t(r_2), \dots, v_t(r_N))$ the wind field on the area R_0 at time t , with $u_t(r_i)$ (resp. $v_t(r_i)$) the zonal (resp. meridional) component at point r_i and time t . More generally, if R represents an arbitrary set of points, $Z_t(R)$ will denote the wind field on this area. In sub-Section 2.2 we propose a model for the multivariate time series $\{Z_t(R_0)\}_{t \in \mathbb{N}}$ with values in $\mathbf{R}^{70}$.

It is well known that meteorological time series are generally non-stationary in time, with seasonal and daily components, and eventually a trend. In this article, we only consider the data corresponding

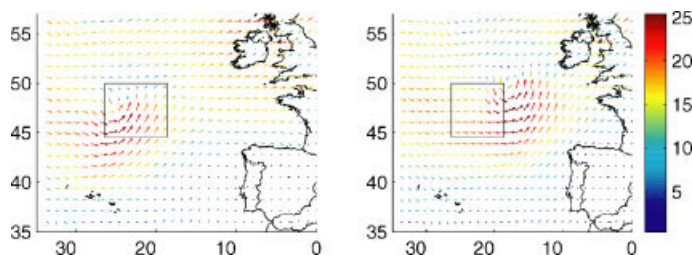


Figure 1. Wind fields on 4 January 1997: 12:00 on the left and 18:00 on the right

to the month of January. We did not find any daily components or trend in this time series, and we will assume that the 11 months of January that are available in our dataset are independent realizations of a stationary and ergodic process.

2.2. Model description

Wind conditions in the studied area are closely linked to the position and to the movement of the principal air masses which govern the climatology of the region (vast low-pressure or high-pressure systems). Depending on these positions, there may be systems of smaller amplitude (fronts or cyclones, for example) that go through the considered area. An example which corresponds to 4 January 1997 is given in Figure 1. At this date, a low pressure goes across the area R_0 between 12:00 and 18:00.

In Boukhanovsky *et al.* (2003a) and Malmberg *et al.* (2004), classical AR models are used in order to describe the space–time evolution of the wind fields. These models can successfully describe the motions of objects that translate at a constant speed. However, the motions of the meteorological structures depend on the position of the principal air masses. As a consequence, these motions evolve in time and AR models are not adapted. In this article, we propose a more elaborate model, in which the motions are introduced explicitly. For this, let us denote as S_t the translation of the wind field $Z_{t-1}(R_0)$ between the dates $t - 1$ and t and as $\mathbf{S} \subset \mathbf{R}^2$ the state space of this process. In order to well define this translation, it is necessary to assume that all the points of the area R_0 move with the same speed, and in particular that the meteorological structures do not rotate and do not change their sizes. This assumption seems physically realistic according to the size of R_0 , and the model proposed in this article could be adapted in order to describe more complex motions than pure translations. In Aberg *et al.* (2002), the evolution of the wind fields is modeled using image warping, and a detailed discussion on the motions of wind fields can be found in this article.

The translations $\{S_t\}$ cannot be observed directly, and thus are introduced as a hidden process. Then, in order to describe the evolution of the wind fields $\{Z_t(R_0)\}$ conditionally to this hidden process, we use linear autoregressive models with time-varying coefficients. More precisely, it is assumed that

$$Z_t(R_0) = A^{(S_t)}Z_{t-1}(R_0) + B^{(S_t)} + H^{(S_t)}E_t \quad (1)$$

where $\{E_t\}$ represents a standard Gaussian noise with 0 mean and identity covariance matrix, $A^{(s)} \in \mathbf{R}^{2N \times 2N}$, $B^{(s)} \in \mathbf{R}^{2N \times 1}$ and $H^{(s)} \in \mathbf{R}^{2N \times 2N}$ are matrices that depend on a reduced number of parameters $\theta \in \Theta$. In (1), $A^{(S_t)}Z_{t-1}(R_0) + B^{(S_t)}$ represents a forecast of the wind field on the area R_0 at time t conditionally to the field on R_0 at time $t - 1$ and to the translation S_t , whereas $H^{(S_t)}E_t$ represents the conditional error. This is illustrated on Figure 2.

Let us now give more details on the modeling of the evolution of the hidden process $\{S_t\}$ and on the parametrization of the matrices $A^{(s)}$, $B^{(s)}$ and $H^{(s)}$. In sub-Section 3.1, we briefly explain how we can check the truthfulness of the different assumptions made hereafter.

2.2.1. Parametrization of the evolution of $\{S_t\}$. First, for the sake of simplicity, we assume that the state space $\mathbf{S}$ of the hidden process $\{S_t\}$ is a finite subset $\{\mathbf{s}_1, \dots, \mathbf{s}_M\}$ of $\mathbf{R}^2$ which corresponds to displacements on the grid of the hindcast model at a speed lower than 40 ms^{-1} . With this choice, the cardinal M of $\mathbf{S}$ is about 300. This assumption makes it possible to compute the maximum likelihood

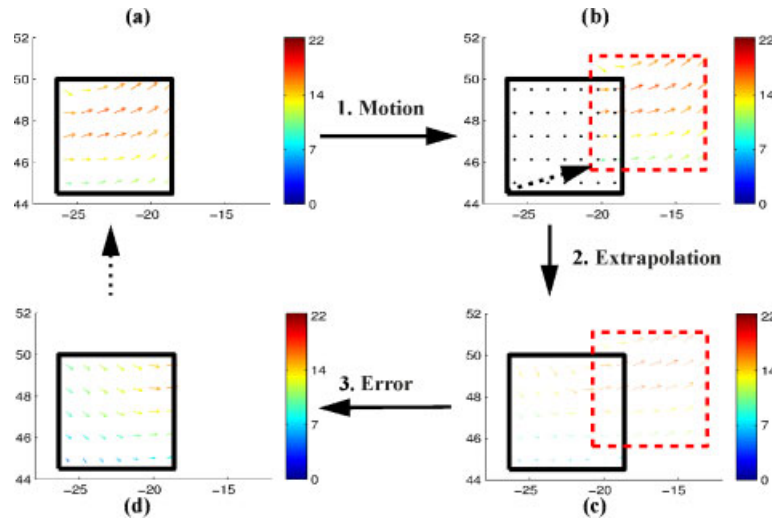


Figure 2. (a) Wind fields on 4 January 1997, 12.00; (b) translated field (dashed box) and initial area R_0 (solid box); (c) translated field (dashed box) and predicted field $A^{(S_t)}Z_{t-1}(R_0) + B^{(S_t)}$ on R_0 (solid box); (d) wind fields on 4 January 1997, 18.00

estimates using standard optimization algorithms (see sub-Section 3.2), and does not seem too restrictive in practice. Indeed, the motions are large compared to the distance between two grid points of the numerical model, so that the error induced by the discretization of the state space is low.

Then, we assume that the hidden process is a first order Markov chain and Q will denote its transition matrix. It is assumed to be a discretized Gaussian kernel: for all $i, j \in \{1 \dots, M\}$,

$$Q(i, j) = P(S_t = s_j | S_{t-1} = s_i) \sim \exp\left(-\|s_i - s_j\|^2 / \sigma^2\right) \exp((s_j - s_0)' \Gamma (s_j - s_0)) \quad (2)$$

with unknown parameters $\sigma > 0$, $s_0 \in \mathbf{R}^2$ and $\Gamma \in \mathbf{R}^{2 \times 2}$ a positive definite symmetric matrix, and $\|\cdot\|$ the Euclidean norm on $\mathbf{R}^2$.

2.2.2. Parametrization of $A^{(s)}$. The matrices $A^{(s)}$ are chosen such that they perform linear extrapolation of the field on the area R_0 from the field on $R_0 + s$. This choice seems natural since, by definition of S_t , we know that if $S_t = s$ then

$$Z_{t-1}(R_0) = Z_t(R_0 + s) + \delta_t$$

with δ_t the deformation of the wind field $Z_{t-1}(R_0)$ between dates $t-1$ and t .

More precisely, if X and Y are two random vectors, let us denote by $\text{cov}(X, Y) = E[XY'] - E[X]E[Y']$ the covariance matrix of X and Y . The matrices $A^{(s)}$ are then defined as

$$A^{(s)} = \text{cov}(Z_t(R_0), Z_t(R_0 + s))(\text{cov}(Z_t(R_0), Z_t(R_0)) + D)^{-1} \quad (3)$$

where $D = \text{cov}(\delta_t, \delta_t)$ represents the covariance matrix of the deformation, which is assumed to be independent of $\{S_t\}$. Equation (3) comes from classical least-squares estimates in linear regression

models. In practice, the spatial covariance matrices $\text{cov}(Z_t(R_0), Z_t(R_0 + s))$ and $\text{cov}(Z_t(R_0), Z_t(R_0))$ are estimated directly from the data, thanks to our assumption of temporal ergodicity. For example, the estimate of $\text{cov}(Z_t(R_0), Z_t(R_0 + s))$ we have used is

$$\hat{\text{cov}}(Z_t(R_0), Z_t(R_0 + s)) = \frac{1}{T} \sum_{t=1}^T z_t(R_0) z_t(R_0 + s)' - \frac{1}{T} \sum_{t=1}^T z_t(R_0) \frac{1}{T} \sum_{t=1}^T z_t(R_0 + s)'$$

where $\{z_t\}_{t \in \{1 \dots T\}}$ denotes the time series of observed wind fields. The estimation of the matrix D is discussed in Section 3.1.

2.2.3. Parametrization of $B^{(s)}$. We can check that the spectral radius of the matrices $A^{(s)}$ defined above are lower than 1. It implies that the AR(1) model defined as

$$Y_t = A^{(s)} Y_{t-1} + B^{(s)} + H^{(s)} E_t$$

admits a unique stationary distribution. Let us denote $M^{(s)} = (I - A^{(s)})^{-1} B^{(s)}$ as the mean of this stationary distribution. $M^{(s)}$ strongly depends on s since the mean wind in a meteorological structure is related to its speed and direction of displacement. To model this relation, we use the linear model described hereafter. Let us denote $M^{(s)} = (m_u^{(s)}(1), \dots, m_u^{(s)}(N), m_v^{(s)}(1), \dots, m_v^{(s)}(N))'$ with $m_u^{(s)}(i)$ (resp. $m_v^{(s)}(i)$) as the mean of the zonal (resp. meridional) component at point r_i . We assume that:

1. $m_u^{(s)}(i) = m_u^{(s)}$ and $m_v^{(s)}(i) = m_v^{(s)}$ for all $s \in \mathbf{S}$ and $i \in \{1 \dots N\}$. In other terms, it is assumed that the mean of the stationary distribution is the same at the different locations in R_0 .
2. $m^{(s)} = Fs + G$ for all $s \in \mathbf{S}$ with $m^{(s)} = (m_u^{(s)}, m_v^{(s)})'$, $F \in \mathbf{R}^{2 \times 2}$ and $G \in \mathbf{R}^{2 \times 1}$. Here, F and G denote matrices of unknown parameters.

2.2.4. Parametrization of $\Sigma^{(s)} = H^{(s)} \times (H^{(s)})'$. The forecast error $Z_t(R_0) - A^{(s_t)} Z_{t-1}(R_0) - B^{(s_t)}$ is mainly due to two terms:

- the deformation of the wind field $Z_{t-1}(R_0)$ between dates $t - 1$ and t
- the extrapolation error which is due to the variability of the shape of the meteorological structures and the arrival of new structures coming in R_0 .

In order to model these two terms, we make the following assumptions on the covariance matrices of the error $\Sigma^{(s)}$:

1. First, in order to model the diagonal of $\Sigma^{(s)}$, we suppose that:

$$\begin{aligned} \Sigma^{(s)}(i, i) &= f_u(\text{dist}(r_i, R_0 + s)) \\ \Sigma^{(s)}(i + N, i + N) &= f_v(\text{dist}(r_i, R_0 + s)) \end{aligned} \tag{4}$$

for all $s \in \mathbf{S}$ and $i \in \{1 \dots N\}$. In other terms, it is assumed that the variance of the error on the zonal (resp. meridional) component at a point r_i , namely $\Sigma^{(s)}(i, i)$ (resp. $\Sigma^{(s)}(i + N, i + N)$), only depends on the Euclidean distance $\text{dist}(r_i, R_0 + s)$ between this point and the translated area $R_0 + s$. When this distance is null, the error is only due to the deformation of the wind fields, and then the error increases with the distance as the extrapolation error increases. Several parametric forms can

be chosen for f_u and f_v . In this work, we have used $f_u(d) = \beta_u + \alpha_u \exp(-d/d_u)$ and $f_v(d) = \beta_v + \alpha_v \exp(-d/d_v)$, where β_u , α_u , d_u , β_v , α_v and d_v are unknown positive parameters.

2. Then, it remains to model the correlation matrix of the error $R^{(s)}$, defined as

$$R^{(s)}(i, j) = \Sigma^{(s)}(i, j) / \sqrt{\Sigma^{(s)}(i, i) \times \Sigma^{(s)}(j, j)}$$

For this, we assume that, for all $s \in \mathbf{S}$:

$$\begin{aligned} R^{(s)}(i, j) &= g_u(\|r_i - r_j\|) \quad \forall (i, j) \in \{1, \dots, N\}^2 \\ R^{(s)}(i + N, j + N) &= g_v(\|r_i - r_j\|) \quad \forall (i, j) \in \{1, \dots, N\}^2 \\ R^{(s)}(i, j) &= 0 \quad \forall (i, j) \in \{1, \dots, N\} \times \{N + 1, \dots, 2N\} \end{aligned} \quad (5)$$

It means that the correlation between the errors on the zonal (resp. meridional) components at two points r_i and r_j only depends on the Euclidean distance $\|r_i - r_j\|$ between these points, and that there is no correlation between the error on the zonal and meridional components. The functions g_u and g_v must satisfy $g_u(0) = 1$ and $g_v(0) = 1$ and may be decreasing. In practice, we assume that $g_u(d) = \exp(-d/l_u)$ and $g_v(d) = \exp(-d/l_v)$, where l_u and l_v are unknown positive parameters. With this parametrization, the matrices $R^{(s)}$ and thus $\Sigma^{(s)}$ are positive definite symmetric (see Cressie, 1993, p. 85).

The proposed model belongs to the family of the Markov Switching AutoRegressive models (denoted MS-AR hereafter). This kind of model has first been introduced by Hamilton in 1989 (see Hamilton, 1989) to analyze the rate of growth of U.S.A. GNP and then used in different fields of applications to model time series subject to regime shifts. The proposed MS-AR model has 20 parameters: 6 for the matrix Q , 6 for the matrices $B^{(s)}$ and 8 for the matrices $\Sigma^{(s)}$. We will denote by $\theta \in \Theta \subset \mathbf{R}^{20}$ the set of unknown parameters. The estimation of these parameters is described in the next section.

3. ESTIMATION

The most usual approach to estimate the parameters of an MS-AR model is to compute the maximum likelihood estimates (MLEs). The numerical computation of the MLE in models with hidden variables has been addressed by many authors. The most popular method is probably the EM algorithm, which was first introduced by Baum *et al.* (1970) for Hidden Markov Models (HMM) and then generalized to other models with missing variables by Dempster *et al.* (1977). However, this algorithm has several well-known limitations, one of them being the possible convergence to local maxima of the likelihood function. It is then important to use a good guess of the MLE as the starting point of the algorithm.

In order to get a first estimate of the unknown parameters, a natural method consists in estimating the values taken by the hidden process $\{S_t\}$ and then using this estimated sequence in order to compute empirical estimates of the quantities of interest. We have used such an approach in this work, and the proposed method is described in sub-Section 3.1. Then, in sub-Section 3.2, we briefly describe the Generalized EM algorithm, which has been used to compute the MLE, and we discuss some statistical properties of these estimates.

3.1. Preliminary estimation

Let us first describe how we have estimated the hidden translations $\{S_t\}$. For this, we have used extra information which is available in the dataset. Indeed, it describes the wind conditions not only on R_0 , but everywhere on the earth, so that it is possible to compute

$$(\hat{s}_2, \dots, \hat{s}_T) = \operatorname{argmax}\{f(s_2, \dots, s_T) | (s_2, \dots, s_T) \in \mathbf{S}^{T-1}\}$$

with

$$f(s_2, \dots, s_T) = \prod_{t=2}^T \exp\left(-\frac{\|z_{t-1}(R_0) - z_t(R_0 + s_t)\|^2}{d_0^2}\right) \times Q(s_{t-1}, s_t) \quad (6)$$

In (6), Q denotes the transition matrix defined in (2) and d_0 a fixed real number. The first term in (6) constrains the estimated sequence of displacements $\{\hat{s}_t\}$ to be such that $z_{t-1}(R_0) \approx z_t(R_0 + \hat{s}_t)$, and the second term will penalize sequences which evolve too quickly in time and thus act as a smoothing term. The function f can be interpreted as the likelihood function of an HMM, and thus the sequence $(\hat{s}_2, \dots, \hat{s}_T)$ can be computed quickly using the Viterbi algorithm (a recent review on HMM can be found in Ephraim and Merhav, 2002).

The choice of the function f is arbitrary, and there exists no objective criterion to validate this choice. In order to check the meteorological realism of the estimated sequences, we performed visual verification. An example is given in Figure 3. It shows that the method succeeds in estimating the movement of the low pressure which goes through the studied area on 4 January 1997. In particular, the two fields represented in the dashed box, and which represent respectively $z_{t-1}(R_0)$ (thick arrows) and $z_t(R_0 + \hat{s}_t)$ (thin arrows), are close to each other. In practice, we have used the numerical values $d_0 = 50 \text{ km}$, $\sigma = 3 \text{ ms}^{-1}$ and $\Gamma = 0$.

Once these displacements have been estimated, we can compute empirical estimates of the covariance matrix D of the deformation which is used to define the matrices $A^{(s)}$ (see equation 3). For this, we have computed

$$\hat{D} = \frac{1}{T-1} \sum_{t=2}^T [z_{t-1}(R_0) - z_t(R_0 + \hat{s}_t)][z_{t-1}(R_0) - z_t(R_0 + \hat{s}_t)]'$$

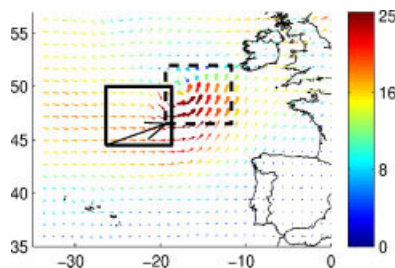


Figure 3. Situation on 4 January 1997 at 18:00. Solid box: R_0 ; dashed box: $R_0 + s$; thin arrows: wind field at 18:00; thick arrows: wind field observed at 12:00 on R_0

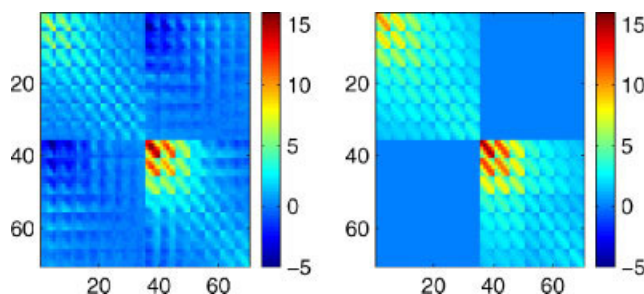


Figure 4. Empirical covariance matrix of the error (displacement of 3 grid steps through the east). Empirical version estimated from the sequence $\{s_t\}$ on the left and parametric version on the right

In the same way, we can use the sequence $\{\hat{s}_t\}$ in order to estimate the conditional means $M^{(s)}$, the covariance matrices of the error $\Sigma^{(s)}$ and the transition matrix Q . These estimates can first be used in the analysis part, in order to guess parsimonious and realistic parametric forms for these matrices and make it possible to check the realism of the different assumptions made in Section 2.2. They can also be used to fit the chosen parametric models by using, for example, least squares estimates. As an example, we have plotted in Figure 4 the empirical covariance matrix of the error corresponding to a displacement s of three grid steps to the east and its parametric version. The shapes of these two matrices are in a good agreement, and thus the chosen parametrization seems to be fitted to our data.

3.2. Maximum likelihood estimates

The estimates obtained in the previous sub-Section have then been used as a starting point to an optimization algorithm. Let us briefly describe this algorithm, which belongs to the family of the Generalized EM algorithms. If θ_{n-1} denotes the value of the parameters after $n - 1$ iterations, the n th iteration consists of two main steps:

- **E-step.** This consists of calculating the same auxiliary function as in the classical EM algorithm (see Dempster *et al.*, 1977). This function is denoted by $R(\theta, \theta_{n-1})$. When the hidden state space $\mathbf{S}$ is finite, it can be computed quickly thanks to the Forward–Backward algorithm which has been first proposed by Chang *et al.* (1966). Otherwise, one can use Monte Carlo approximations of this function (see Douc *et al.* (2004) and references therein).
- **M-step.** In the usual EM algorithm, this step consists in maximizing the function $R(\theta, \theta_{n-1})$, and updating the current value of the parameters as $\theta_n = \operatorname{argmax}_{\theta \in \Theta} R(\theta, \theta_{n-1})$. For the model proposed in Section 2, we did not find any analytical solution to this optimization problem, so we had to use an iterative numerical optimization procedure. When the iterations stop with a non-optimal value θ_n , which is such that $R(\theta_n, \theta_{n-1}) > R(\theta_{n-1}, \theta_{n-1})$, the algorithm is called a Generalized EM algorithm. This algorithm was first proposed by Dempster *et al.* (1977), and general conditions which imply the convergence of this algorithm to a local maximum of the likelihood function can be found in Wu (1983). In practice, we have used some iterations of a quasi-Newton algorithm to compute θ_n . Each iteration of this algorithm is time-consuming because of the complexity of the objective function and the number of parameters. Thus, a careful choice of this number of iterations, denoted by k_n , has to be done. According to the numerical tests that we have done, the best strategy seems to choose a small number of iterations for small values of n and then to increase this number of iterations progressively when the current values of the parameters get closer to a maximum of the likelihood function. In practice, we have used $k_n = n$.

Whereas there exists a considerable amount of literature about the numerical computation of MLE, the statistical issues concerning the asymptotic properties of these estimators have been addressed only recently. Explicit conditions which warrant the consistency of the MLE in MS-AR models have first been established simultaneously in 1998 by Krishnamurthy and Ryden and Francq and Roussignol, and assumptions which guarantee consistency and asymptotic normality of MLE in MS-AR models with compact, not necessary finite, hidden state space $\mathbf{S}$ can be found in Douc *et al.* (2004). It would be beyond the scope of this article to show that the MS-AR model proposed in Section 2 verifies the assumptions given by these authors.

4. MODEL VALIDATION

In order to validate the MS-AR model, we have checked its ability to perform short-term forecasts, and the obtained results have been compared to those of an AR(1) model. For MS-AR models, the best one-step prediction can be expressed from the forecast probabilities $P[S_t = s|Z_1(R_0), \dots, Z_{t-1}(R_0)]$ as

$$\begin{aligned} & E[Z_t(R_0)|Z_1(R_0), \dots, Z_{t-1}(R_0)] \\ &= \sum_{s \in \mathbf{S}} P[S_t = s|Z_1(R_0), \dots, Z_{t-1}(R_0)] E[Z_t(R_0)|Z_{t-1}(R_0), S_t = s] \\ &= \sum_{s \in \mathbf{S}} P[S_t = s|Z_1(R_0), \dots, Z_{t-1}(R_0)] (A^{(s)} Z_{t-1}(R_0) + B^{(s)}) \end{aligned} \quad (7)$$

and the forecast probabilities can be computed recursively (Forward algorithm). The generalization to other forecast horizons is straightforward (see Krolzig, 1997).

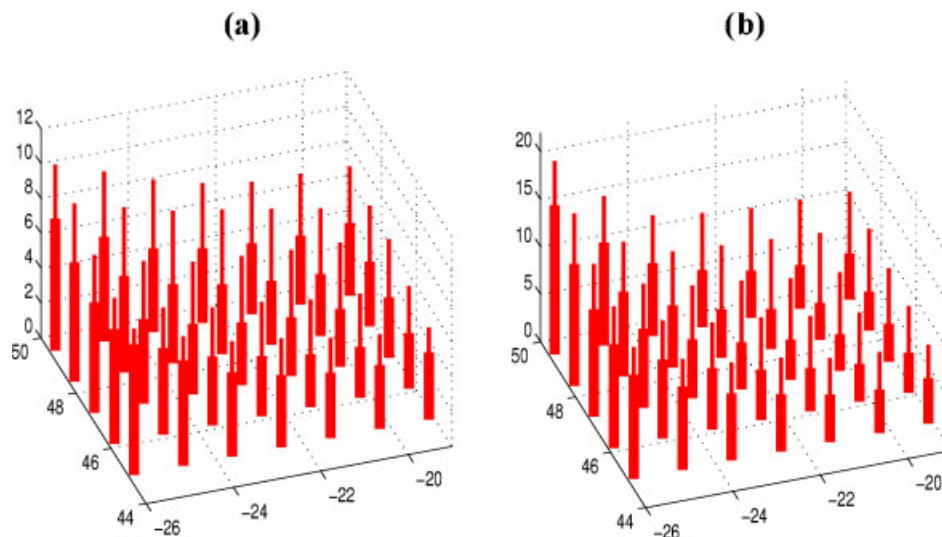


Figure 5. Variance of the one-step-ahead forecast error on the zonal (a) and meridional (b) components at the different points of R_0 (thick line: MS-AR model, thin line: AR(1) model)

Table 1. Evolution of the Perron–Frobenius norms with the forecast horizon h

$h(\text{hour})$	6	12	18	24	30
MS-AR	151	538	952	1263	1498
AR	300	981	1719	2257	2560

In Figure 5, we have plotted the variance of the one-step-ahead forecast error on the zonal and meridional components at the different locations of R_0 , for the MS-AR and AR(1) models. The MS-AR model gives significantly better forecasts for all locations of the domain R_0 , the improvement being more important at the points located at the east of the domain. It is due to the fact that these points are generally closer to the translated area $R_0 + S_t$ since the air-masses generally move toward the east. Thus, at these locations, the extrapolation error is generally low and the error is mainly due to the deformation of the wind fields. The errors at the points located at the west, on the other hand, are more important because the ‘new’ meteorological structures come in the area R_0 from the west.

We have also computed the variance of the error corresponding to other time-step prediction. In order to compare the variance–covariance matrices of the forecast errors corresponding to the AR(1) and the MS-AR models, we have calculated their Perron–Frobenius norms. It is defined, for any matrix Σ , as $\|\Sigma\|_{PF} = \sqrt{\text{tr}(\Sigma'\Sigma)}$. The results are given in Table 1. The MS-AR model clearly improves the results obtained with the AR(1) model.

5. CONCLUSION

In this article, an original Markov Switching Autoregressive model is proposed for modeling the space–time evolution of the wind. The main originality of this model consists in introducing the movement of the air masses as a hidden variable which governs the short-term evolution of the wind fields. The model is fitted on hindcast data in North Atlantic and it is shown that it provides better short-term forecasts than does the AR(1) model.

This model has been developed specifically for wind fields, but similar methodology could be used for other space–time processes in which motions are present. In the proposed model, the state space of the hidden process is assumed to be finite. This assumption seems realistic for wind fields, but restrictive for other space–time processes. The model could be refined by allowing continuous translations or more complex motions, such as rotations, for instance.

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